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Paleomagnetism, paleointensity and geochronology of Miocene basalts and baked sediments from Velay Oriental, French Massif Central

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Abstract. Paleomagnetic study of Miocene basalts and baked sediments from Velay Oriental, France, yields a paleomagnetic pole with coordinates 84.1°N, 171.2°E (A_{95} = 8.6°, K = 29, N = 11). The pole is supported by a positive reversal test, and it corresponds well with the apparent polar wander path for Eurasia for 10 Ma. Paleosecular variation, estimated as the angular standard deviation of the VGP distribution 14.9° +6.5°/-3.5°, is close to expected, suggesting that the paleomagnetic pole represents a time-averaged field. Moreover, five new $^{40}Ar/^{39}Ar$ plateau ages, falling between 9.20 and 13.55 Ma, were obtained. Thellier paleointensity experiments were carried out on both basalts and baked sediments. Almost all basalts are chemically unstable during the laboratory heatings and of the 36 preselected samples only three yield usable results. In contrast, the baked sediments are ideal for Thellier experiments with 11 out of 13 samples giving reliable paleointensity estimates. Rock magnetic experiments show that the main ferromagnetic minerals in the baked sediments are fine-grained magnetite and hematite with a large proportion of superparamagnetic grains. In addition to the standard partial thermoremanent magnetization (pTRM) check, used in Thellier experiments to detect chemical/crystalline alteration, we introduce a second "pTRM tail" check designed to test the basic prerequisite of the method, namely, the equality of unblocking and blocking temperatures. For the baked sediments, unblocking and blocking temperatures are found to be identical, which indicates that the remanence is carried by single-domain grains.

1. Introduction

The Tertiary-Quaternary volcanics of Massif Central constitute an excellent natural laboratory of paleomagnetism, offering the first finding of reverse magnetized rocks [*Brunhes and David,* 1903], some of the earliest magnetostratigraphic studies [e.g., *Roche*, 1951, 1953] and the first discovery of a geomagnetic excursion [*Bonhommet and Babkine*, 1967]. Through the many years of study, a long list of paleomagnetic poles have been obtained for Massif Central. However, the vast majority of poles were obtained prior to the advent of stepwise demagnetization and principal component analysis. Moreover, the ages of the poles are often poorly confined. In this study we present new paleomagnetic results from Miocene outcrops of Massif Central sampled in Velay Oriental accompanied by five precise $^{40}Ar/^{39}Ar$ ages and define a representative paleomagnetic pole. In the following, this paleomagnetic pole is discussed in relation to previous studies of Massif Central and the Eurasian apparent polar wander path.

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The major objective for our sampling was to obtain reliable estimates of the paleofield strength. Among 200 published paleointensity determinations for Miocene, very few fulfill basic minimum reliability criteria [*Perrin and Shcherbakov*, 1997]. Hence, on the basis of the available data set it is hardly possible to draw any conclusions about the strength of the Earth's magnetic field in Miocene times. In this paper we present new Miocene paleointensity estimates obtained from both basalts and baked sediments of Velay Oriental. Special attention is given to the baked sediments and their suitability for Thellier experiments. We discuss the Thellier paleointensity method and suggest introducing a simple additional partial thermoremanent magnetization (pTRM) check in order to identify the domain structure of the remanence carriers.

2. Geology and Sampling

The Tertiary-Quaternary volcanic activity of the French Massif Central is part of a much larger igneous province extending from central Spain through Germany to the Ohre Rift of the Czech Republic and into Poland. This widespread magmatic activity is difficult to link in simple terms to the Tertiary-Quaternary extensional thinning of the western

European lithosphere because the main volcanic areas are located on uplifted Variscan basement massifs, while the N-S trending graben systems show only minor igneous activity [*Wilson and Downes*, 1992]. The asthenospheric mantle source of the primary magmas and the broad uplift of the thermal lithosphere-asthenosphere boundary [*Perrier and Rueg*, 1973] might suggest that the magmatism is linked to mantle plume activity [*Granet et al.*, 1995; *Sobolev et al.*, 1997].

The Velay Province forms the southeastern part of Massif Central and presents two very distinguishable entities. To the west, the Devès basaltic plateau, Plio-Quaternary in age, covers an area of 600 km^2 between the Loire and the Allier River. This volcanism is undifferentiated, Strombolian in type. To the east, the Velay Oriental is formed by a Mio-Pliocene alkaline volcanic chain which underwent very strong differentiation. These basaltic to phonolitic volcanics stretch over a distance of 55 km with more than 180 points of extrusions, of which 100 form phonolitic lava "domes" or plugs giving rise to the characteristic regional landscape.

For this study we sampled 13 lava flows and two dikes of Miocene age from Velay Oriental. At two sites the underlying thin layer (~15 cm) of baked sediments was sampled as well as the lava flows, and in one site (St. Clement 6) only baked sediments were sampled. All samples were drilled directly in the field with a portable gasoline-powered drill, except for two sites, where baked sediments were sampled as oriented blocks and then later drilled in the laboratory. All samples were oriented using both magnetic and Sun compass. Owing to the fragile nature of the baked sediments, some of the blocks broke and therefore lost their orientation. For example, from the four blocks sampled at St. Clement 6, we were able to obtain only a single oriented sample. In the laboratory the cores were subsampled into standard cylindrical specimens (in general, two to four specimens per core). The geographic location of the sites are shown in Figure 1, and geographic coordinates and ages are given in Table 1. The St. Clement lava flows were given the same numbers as in the initial study of these units by *Prévot* [1963]. Hand blocks were selected for geochronology from five sites: Bournac1, Combevieille, Monastier, Faries, and Croix. The details of sampling of the sites Faries and Croix can be found elsewhere [*Riisager*, 1999].

No local tectonic movements have taken place in Velay Oriental since the emplacement of the Miocene lava flows [*Mergoil et al.*, 1993]. This agrees well with our field observations that all sampled lava flows were horizontal. Consequently, no tilt correction was applied in this study.

3. Geochronology

The single whole rock fragments used for the experiments were carefully handpicked under a binocular microscope from crushed 0.5-mm-thick rock slabs. The samples were wrapped in

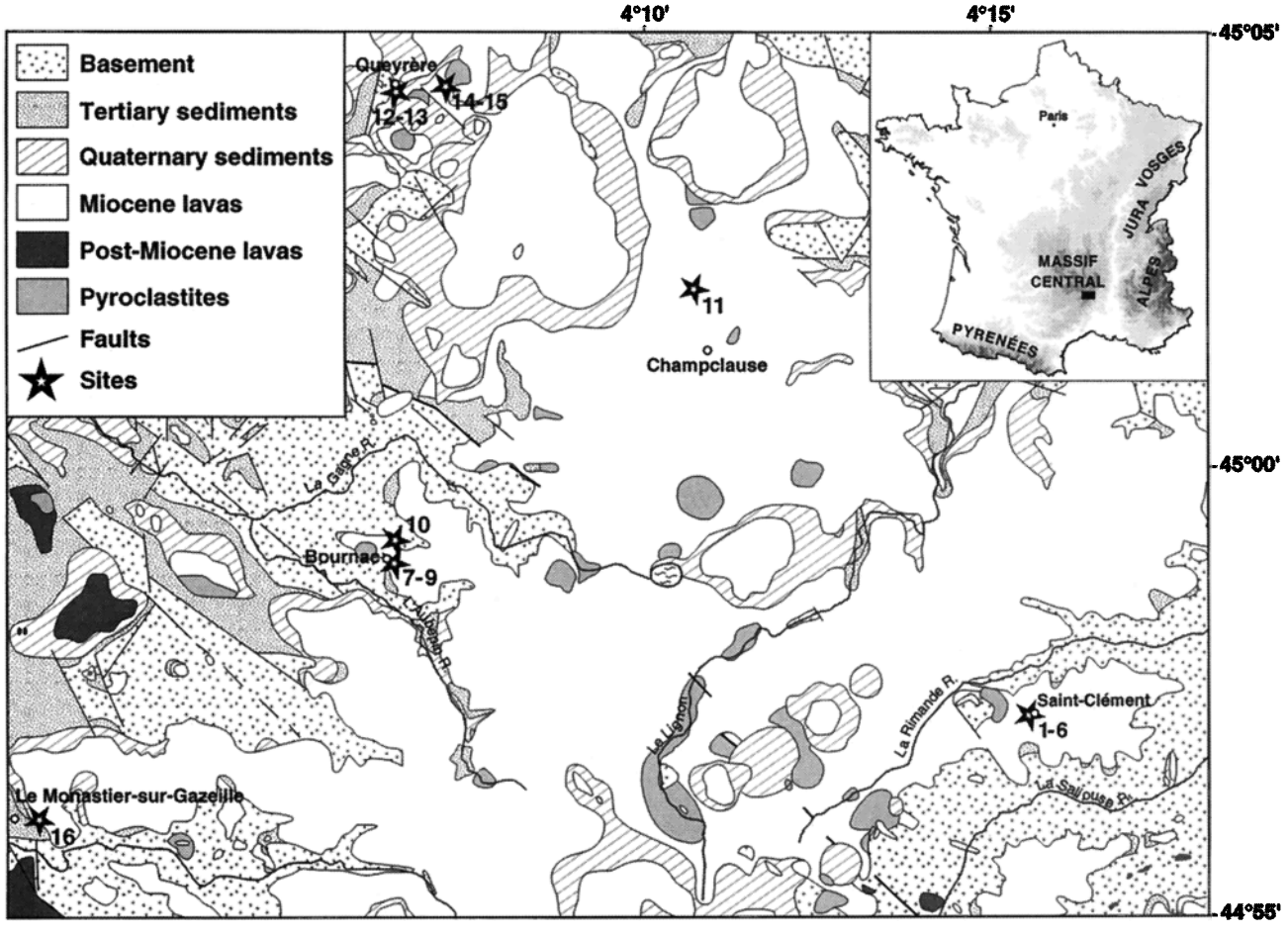


Figure 1. Simplified geological map of Velay Oriental. The locations of the sites sampled in this study are shown with stars. The site numbers are the same as in Table 1.

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Table 1. Paleodirectional Data of Basalts and Baked Sediments From Velay Oriental

No.	Site	Lat, °N	Long, °E	Alt, m	Age, Ma	*n/N*	*I*, deg	*D*, deg	*k*	α_{95}, deg	VGP Lat, °N	VGP Long, °E
1	St. Clement 1	44 95	4 28	955	8 - 10	4/5	64.5	10 3	262	5.7	82.7	79 8
2	St. Clement 4	~	~	975	~	5/6	48.0	60.8	36	12.9	transition?	
3	St Clement 5	~	~	985	~	4/5	-55 1	208.7	296	5.3	-66 4	287.5
4	St. Clement 6	~	~	995	~	1/1	-60 7	144 8				
5	St. Clement 7	~	~	1005	~	3/5	-51 3	177 8	927	4 1	-76.9	12.5
6	St. Clement 8	~	~	1105	~	3/5	-3 9	202.6	57	16.5	viscous	
7	*Bournac 1	44.98	4.10	1170	9.38 ± 0.17[b]	7/7	66.5	336.5	123	5.5	73.5	296.6
	Baked sediments					2/2	74.4	350.4				
	Lava flow					5/5	63 1	333 3	349	4.1		
8	Bournac 2	44.98	4.10	1080	9.7 ± 1.0	5/5	-37 4	174.3	153	6.2	-65.5	17.0
9	Crouzet	44 97	4.07	1100	8 - 10	10/10	58.9	352 4	109	4.6	82.3	233.3
10	Bigorre	45.00	4.10	1180	10	0/5					lightning	
11	Faurie	45.04	4.18	1200	7.95 ± 0.15	11/11	53.4	347.8	165	3.6	75.5	228.7
12	Queyrières (dike)	45.07	4.10	1150	11 2	0/6					lightning	
13	Valpiliere (dike)	45.08	4 10	1150	10.3	4/5	-61 7	193.9	641	3 6	-79.8	281 6
14	*Combevieille	45.07	4 15	1150	13.55 ± 0.05[b]	10/10	70 5	12 3	543	2.1	77.6	39.0
	Baked sediments					5/5	71.2	15.8	1446	2.0		
	Lava flow					5/5	69.7	9.1	361	4.0		
15	Vestias	45.08	4.15	1020	8 - 10	4/5	49.1	352.9	374	4.8	73 9	206.9
16	Monastier	44.93	4.02	950	9.20 ± 0.10[b]	6/6	56.7	14.8	587	2.8	76.5	123.4

Means are latitude, 84.1°, longitude, 71.2°; $K = 29$, $A_{95} = 8.6°$, and $N = 11$.

Lat, Long are the geographic coordinates of sampled sites; Alt, m is the altitude of the site in meters; Age, Ma is K/Ar or stratigraphic ages [*Mergoil et al.*, 1993]; *N* is number of treated samples; *n* is number of samples used for calculation; *I*, *D* are the inclination and declination of the site-mean ChRMs; *k*, α_{95} are the precision parameter and the confidence cone of the Fisher statistics [*Fisher et al.*, 1987]; VGP (Lat, Long) are the latitude and longitude of the virtual geomagnetic pole.

*Sites where site-mean ChRM was obtained on both basalts and baked sediments, with the mean directions calculated separately for the basalts and baked sediments.

[b]The $^{40}Ar/^{39}Ar$ ages (Table 2)

Al foil to form small packets (11 x 11 mm). These packets were stacked up to form a pile within which packets of flux monitors were inserted every 8 to 10 samples. The stack, put in an irradiation can, was irradiated for 5 hours at the McMaster reactor (Hamilton, Canada) with a total flux of 6.4 x 10^{17} n cm^{-2}. The irradiation standard was the sanidine Fish Canyon (28.02 Ma [*Renne et al.*, 1998]). The sample arrangement allows us to monitor the flux gradient with a precision as low as ±0.2%.

The step-heating experiment procedure was described in detail by *Ruffet et al.* [1991, 1995]. The whole rock fragments analyses are performed with a laser probe using a continuous argon ion laser model. Blanks are performed routinely each first or third run and are subtracted from the subsequent sample gas fractions. Typical blank values were in the range 4.6 x 10^{-13}< M/e40 <6.2 x 10^{-13}, 4.3 x 10^{-14}< M/e39 <1.2 x 10^{-13}, 1.1 x 10^{-13}< M/e37 <1.4 x 10^{-13}, and 6.0 x 10^{-14}< M/e36 <7.4 x 10^{-14} cm^{-3} STP.

To define a plateau age, we need a minimum of three consecutive steps, corresponding to a minimum of 70% of the total $^{39}Ar_K$ released, and the individual fraction ages should agree within 2σ with the integrated age of the plateau segment. The ^{39}Ar-^{40}Ar results in Table 2 are displayed at the 1σ level whereas plateau ages are shown at the 2σ level on Figure 2.

The five whole rock samples display rather flat age spectra which allow to define five plateau ages between 9.20 ± 0.10 Ma and 13.55 ± 0.05 Ma (Figure 2). An intermediate temperature step from sample Croix displays an anomalously low apparent age which was related to magnetic field instabilities of the mass spectrometer magnet sector. This apparent age is later ignored. Generally speaking, all the age spectra display integrated ages in good agreement with plateau ages, confirming their high quality. This probably results of combined effects of rather high K contents (and low $^{37}Ar_{Ca}/^{39}Ar_K$ measured ratios [$CaO/K_2O=^{37}Ar_{Ca}/^{39}Ar_K$ x 2.179]) and low atmospheric contaminations, except for Croix sample, which yields the highest Ca/K ratios, atmospheric contamination, and consequently error bars.

4. Rock Magnetic Properties

The rock magnetic experiments include measurements of the temperature dependence of weak field susceptibility, κ(T), in vacuum (better than 10^{-3} mbar) using a Bartington MS2 and hysteresis measurements performed on a Micromag alternating gradient force magnetometer. Also the magnetic viscosity index [*Thellier and Thellier*, 1944], Königsberger ratio, and frequency dependence of magnetic susceptibility ($\chi_{fd\%}$) [*Bloemendal et al.*, 1985] were determined.

On the basis of the rock magnetic experiments the basalt samples can be divided into two major groups: samples with Curie temperatures T_c between 75°C and 150°C (later called "low T_c basalt" for simplicity) and samples with T_c between 450°C and 560°C (high T_c basalt).

The two dikes (sites 12 and 13, Table 1) and most of the lava flows (sites 1-3, 6, 8-9, 11, and 15-16, Table 1) are low T_c basalts. The low Curie temperatures, 75°C to 150°C, indicate that the main magnetic carrier is titanomagnetite with Ti content, $x \approx 0.6$-0.7 [*O'Reilly*, 1976]. The hysteresis loops have rather low coercivities, $B_c \approx$ 5-10 mT, and ratios of saturation remanent magnetization (J_{rs}) over saturation magnetization (J_s) between 0.1 and 0.4, indicating a mixture of single-domain (SD), pseudo-single-domain (PSD) and multidomain (MD) grain sizes [*Day et al.*, 1977]. Some low T_c basalts have a viscosity index >10% and Königsberger ratio <2, indicating that they might not be stable recorders of the paleofield. An example of a low T_c basalt is shown in Figure 3a, having a κ(T) curve that is reversible in repeated heatings. However, a majority of samples show some chemical transformation and less reversible κ(T) curves.

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Table 2. The $^{40}Ar/^{39}Ar$ Analytical Data

Step	$^{40}Ar_{Atm}$, %	$^{39}Ar_K$, %	$^{37}Ar_{Ca}/^{39}Ar_K$	$^{40}Ar^*/^{39}Ar_K$	Age, Ma
		Bournac 1 (RT) J = 0.0011326			
1	71.3	6 1	1.27	5.65	11.50 ± 0.64
2	50.1	5.8	1.06	4.16	8.48 ± 0.54
3	23.5	13.5	0.77	4.72	9.61 ± 0.25
4	10.6	9.5	0.82	4.53	9.23 ± 0.27
6	12.7	8.7	0.74	4.50	9.16 ± 0.32
7	3.1	9.3	1.05	4.90	9.99 ± 0.28
8	3.8	11.9	1.69	4.59	9.35 ± 0.27
Fuse	9.5	35.3	7.59	4.54	9.25 ± 0.10
					9.46 ± 0.09*
		Combevieille (RT) J = 0.0011390			
1	69.6	0.9	1.21	6.11	12.52 ± 1.03
2	24.4	2.8	0.56	6.84	13.99 ± 0.28
3	13.5	5.5	0.34	6.63	13.56 ± 0.15
4	2.7	14.2	0.24	6.71	13.73 ± 0.07
5	1.2	11 0	0.25	6.69	13.69 ± 0.06
6	1.2	11.7	0.24	6.65	13.62 ± 0.07
7	2.2	11.1	0.27	6.55	13.40 ± 0.08
8	1.1	6.1	0.41	6.63	13.57 ± 0.13
9	2.0	3.1	1.11	6.66	13.63 ± 0.23
Fuse	2.9	33.5	2.93	6.56	13.43 ± 0.04
					13.55 ± 0.03*
		Monastier (RT) J = 0.0011306			
1	57.8	13 2	0.16	4.35	8.86 ± 0.22
2	40.3	7.6	0.30	4.42	8.99 ± 0.25
3	34.1	7.4	0.30	4.53	9.21 ± 0.23
4	44.7	7.0	0.38	4.47	9.09 ± 0.16
5	26.1	10.7	0.46	4.67	9.50 ± 0.10
6	18.3	7.0	0.48	4.55	9.25 ± 0.16
7	22.5	5.9	0.71	4.27	8.69 ± 0.25
Fuse	18.4	41 3	4.05	4.53	9.21 ± 0.07
					9.14 ± 0.05*
		Croix (RT) J = 0.0011371			
1	96.1	1.4	1.70	1.90	3.89 ± 2.54
2	73.2	4.6	1.41	4.76	9 75 ± 0.72
3	83.0	11.5	1.33	5.85	11.95 ± 0.52
4	78.3	12 8	1.16	5.48	11.20 ± 1.13
5	26.3	8.8	1.36	6.16	12.58 ± 0.41
6	40 6	12.7	1.12	4.60	9.42 ± 0.30
7	14.6	15.9	0.96	6.06	12.38 ± 0 29
8	36.3	7.0	7.90	5.69	11.63 ± 0.47
Fuse	36.1	25.5	25 46	5.72	11.70 ± 0.20
					11.36 ± 0.19*
		Faries (RT) J = 0.0011497			
1	72.4	5.2	0.03	4.95	10 25 ± 0.61
2	8.0	22.3	0.02	5.95	12.30 ± 0 10
3	5 2	18.7	0.02	5.96	12.32 ± 0 13
4	3.5	13.4	0.02	6.04	12.48 ±.0.22
5	1.0	14.5	0.03	6 15	12 72 ± 0.16
6	2.5	8.5	0.07	6.15	12.72 ± 0.29
Fuse	5.4	17.5	0.23	6.09	12.59 ± 0.13
					12.37 ± 0.07*

The $^{40}Ar_{atm}$ is the atmospheric ^{40}Ar; $^{40}Ar^*$ is the radiogenic ^{40}Ar; Ca, produced by Ca-neutron interferences; *K*, produced by K-neutron interferences; Age, Ma, is the date calculated using the decay constants recommended by *Steiger and Jäger* [1977] The errors are at the 1s level and do not include the error in the value of the *J* parameter. Correction factors for interfering isotopes produced by neutron irradiation in the McMaster reactor were $(^{39}Ar/^{37}Ar)_{Ca} = 7.06x10^{-4}$, $(^{36}Ar/^{37}Ar)_{Ca} = 2.79x10^{-4}$, and $(^{40}Ar/^{39}Ar)_K = 2.95x10^{-2}$.

*Integrated age

The high T_c basalts (sites 5, 7, and 14, Table 1) are characterized by elevated Curie temperatures, 450-560°C, indicating titanomagnetite with low Ti content as the main magnetic carrier. The low Ti content probably results from oxyexsolution of primary titanomagnetites (i.e., low T_c basalts) at high temperature during the initial cooling of the lava. The hysteresis loops indicate medium coercivities, $B_c \approx$ 10-30 mT and PSD grain sizes (Figure 3b). All samples of this group have viscosity indexes <5% and Königsberger ratios >4. An example of a high T_c basalt is shown in Figure 3b, having a κ(T) curve that is nearly reversible up to 450°C, while irreversible at higher temperatures. Many high T_c basalts are characterized by the presence of maghemite, evidenced by irreversible destruction of a magnetic phase at 300°C during κ(T) experiments.

A third particular group of samples is constituted by the baked sediments (sites 4, 7, and 14, Table 1). A typical example is shown in Figure 3c having a markedly wasp-waisted hysteresis loop indicating the presence of at least two distinct coercivity populations [*Roberts et al.*, 1995; *Tauxe et al.*, 1996]. One of these populations is a high-coercivity mineral, probably hematite, evidenced by the hysteresis loop remaining open until high fields (~1.3 T) are reached. Alternating field (AF) demagnetization

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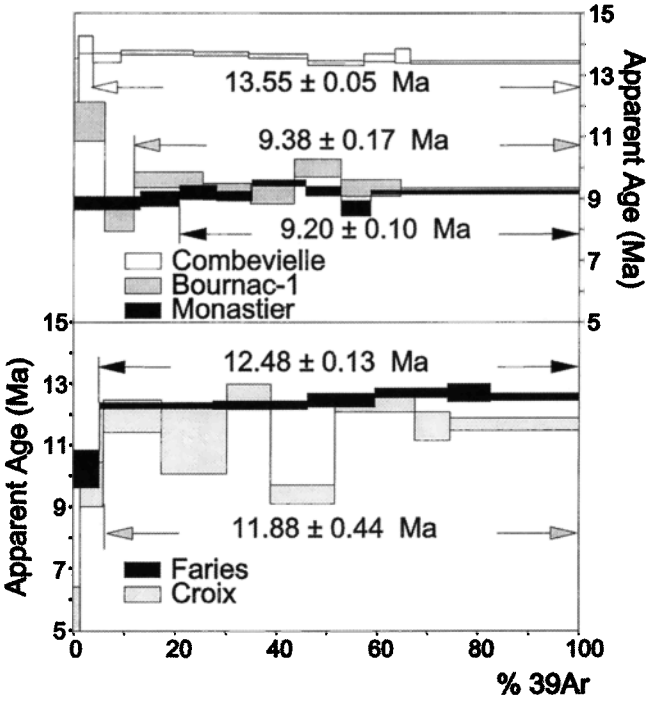


Figure 2. The $^{40}Ar/^{39}Ar$ age spectra yielded by argon ion laser probe analyses of whole rock samples. The age error bars for each temperature steps are at the 1σ level. The errors in the *J* values are not included. Plateau ages error bars are at the 2σ level.

(Figure 4d) also indicates the presence of high-coercivity material with B_c higher than 150 mT. The high natural remanent magnetization (NRM) and J_{rs} intensities of some of the baked sediment samples indicate that a strongly magnetic mineral, probably magnetite, must also be present and in some cases dominate the remanence. Finally, a significant amount of superparamagnetic (SP) particles is suggested by the anomalously high susceptibility, resulting in very low Königsberger ratios and the strong frequency dependence of magnetic susceptibility, with values of $\chi_{fd\%}$ between 8 and 13 % [*Bloemendal et al.*, 1985]. The distinctive 1/T shape of the κ(T) curve (Figure 3c) indicates that the susceptibility is almost entirely governed by SP particles. The fact that no spontaneous magnetization is observed in the κ(T) experiments above 600°C suggests that the dominant SP magnetic mineral is magnetite.

5. Paleodirectional Results, Paleosecular Variation, and Paleomagnetic Pole

Three to five samples per site were subjected to thermal cleaning in vacuum (>10^{-4} mbar), with the initial susceptibility κ measured after each temperature step. An additional two to three samples per site were subjected to alternating field (AF) demagnetization using a laboratory built three-axis AF demagnetizer. Remanences were measured in a magnetic shielded room with a two-axis CTF cryogenic magnetometer or a JR5a spinner magnetometer. The direction of the characteristic remanent magnetization (ChRM) was calculated using standard principal component analysis [*Kirschvink*, 1980]. The acceptable maximum angle of deviation was <3°, and minimum number of demagnetization points was 5.

The demagnetization results follow closely the observations from rock magnetic experiments. The low T_c basalts have low median destructive fields around 10-15 mT and unblocking temperatures T_{ub} lying between 100°C and 200°C. The thermal demagnetization shown in Figure 4a is typical for low T_c basalts, offering an accurate definition of the ChRM direction. Only samples from St. Clement 8 with high viscosity index showed more noisy demagnetization results, making it hard to isolate the directions of ChRM and resulting in larger maximum angular deviations [*Kirschvink*, 1980]. Consequently, the site-mean direction is less well defined for this unit, as also evidenced by the high confidence limit (α_{95}) and low precision parameter (*k*) (see Table 1). The initial susceptibility κ is in many cases changing drastically at higher temperatures, indicating chemical changes of the samples (Figure 4a).

The high T_c basalts have simple univectorial demagnetizations (Figure 4b). Median destructive fields fall in the range 30-50 mT. The susceptibility often undergoes changes throughout the thermal demagnetization, indicating that some demagnetization is probably related to destruction of magnetic mineral rather than unblocking.

The baked sediments showed ideal one-component NRMs in both thermal and AF demagnetizations. The thermal demagnetization is characterized by a wide spectra of unblocking temperatures continuing above 600°C and no significant changes in κ (Figure 4c). AF demagnetization was less effective due to the remanence partly residing in a high-coercive magnetic mineral (hematite) (Figure 4d). It is important to note that when both baked sediments and the overlying "baking" lavas were sampled, the directions for both materials are statistically indistinguishable when both ChRM directions could be well defined (site 14, Table 1). In the case of site 7 the larger difference between the directions might relate to the insufficient number of baked sediments samples.

Two sites (Queyrières and Bigorre) had highly scattered NRM directions and demagnetization results, suggesting a lightning induced isothermal overprint. It was not possible to isolate the primary magnetization for these two sites (see Table 1). In the further analysis of site-mean directions, we chose to exclude the highly viscous St. Clement 8 site, and St. Clement 6 with only one oriented sample. The site-mean direction of St. Clement 4 is lying ~30° away from the mean direction of all other sites, which is on the limit of paleosecular variation, and as this flow is situated between flows with normal and reverse polarities, it might be recording a transition, as has already been suggested by *Prévot* [1975]. We chose to also exclude this site from further analysis.

The site-mean directions of the remaining 11 sites are given in Table 1 and shown in Figure 5. The angle between the mean normal and reverse polarity directions is 9.8°. Performing the reversal test using simulation [*McFadden and McElhinny*, 1990], we find the critical angle, for which the two directional data set would be different at the 95% significance level to be 16.4°. Thus the collection passes the reversal test at level C [*McFadden and McElhinny*, 1990].

In order to evaluate paleosecular variation (PSV) we calculated the angular standard deviation (ASD) of the VGP distribution. Following *McFadden et al.* [1991], ASD was corrected for within-site dispersion, S_w = 4.4°, and found to be 14.9° with 95% confidence limit +6.5°/-3.5° [*Cox*, 1969]. This value of ASD is somewhat lower than predicted (~17°) by the PSV model of *McFadden et al.* [1991] for the given age and paleolatitude. The difference is, however, insignificant,

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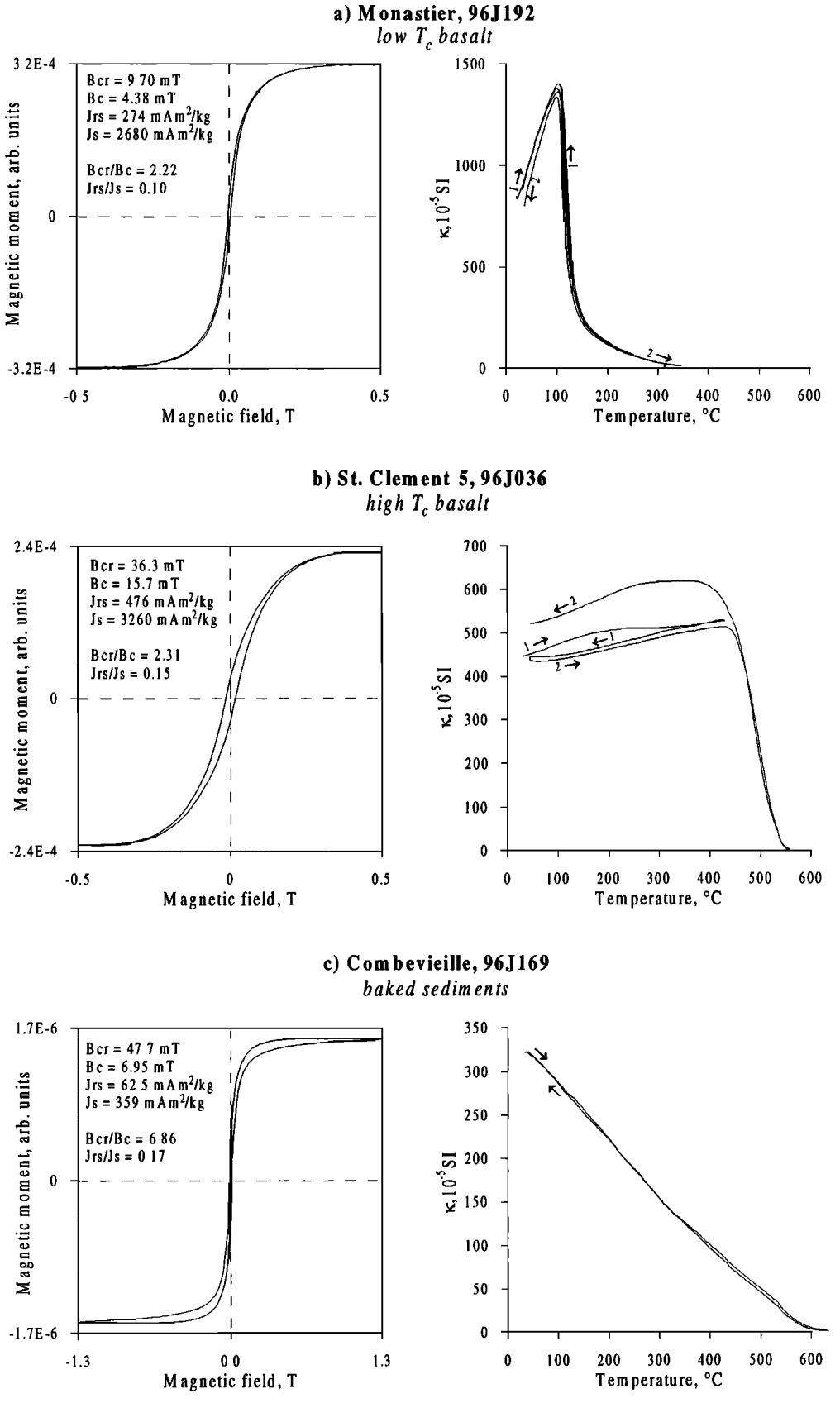


Figure 3. Typical examples of hysteresis loops and susceptibility versus temperature curves for the three groups of samples (see text): (a) low T_c basalts, (b) high T_c basalts, and (c) baked sediments.

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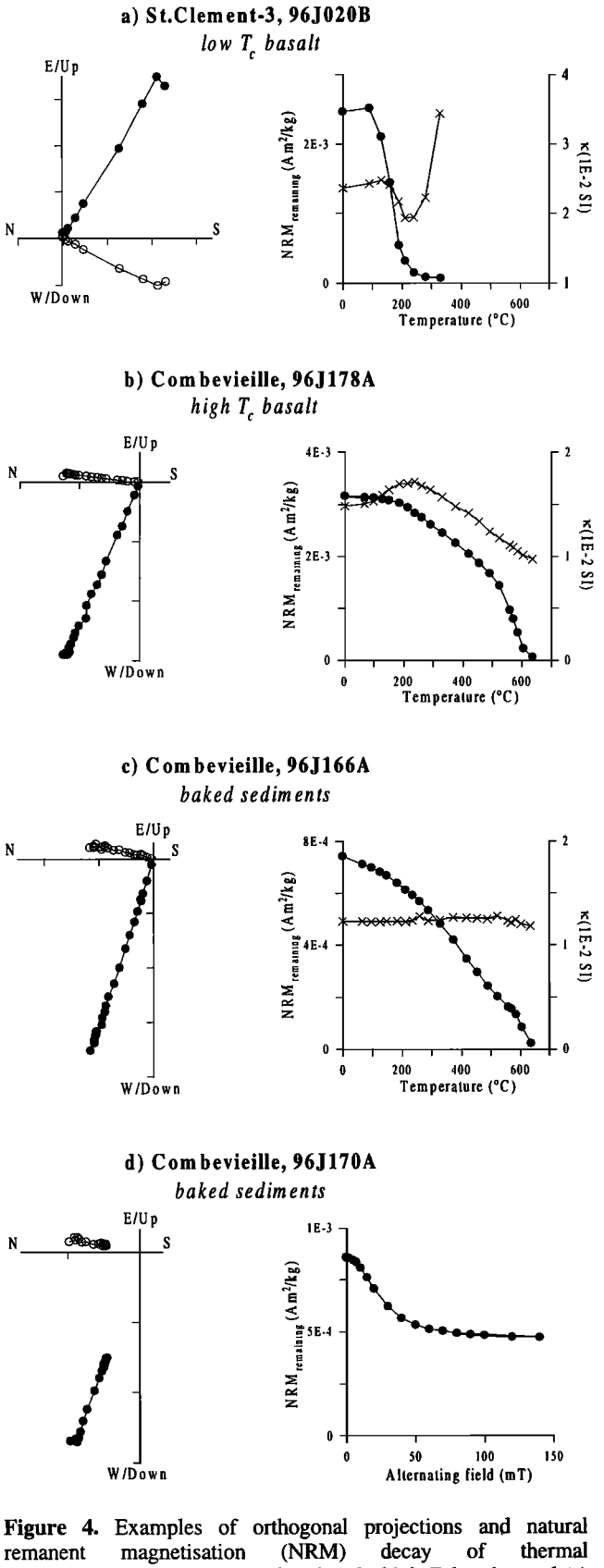


Figure 4. Examples of orthogonal projections and natural remanent magnetisation (NRM) decay of thermal demagnetization of (a) low T_c basalts, (b) high T_c basalts, and (c) baked sediments. Susceptibility κ, measured after each temperature step, is shown by crosses. (d) Example of AF demagnetization of baked sediments. Solid (open) symbols on orthogonal projection correspond to vertical (horizontal) planes.

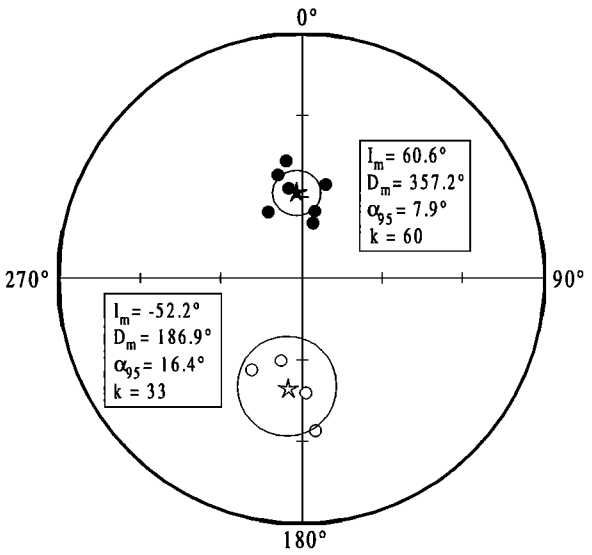


Figure 5. Equal-area projection of mean directions for the 11 sites. Open (solid) symbols correspond to negative (positive) inclination. The mean for each polarity is shown by a star, and the 95 % circles of confidence are also given.

suggesting that secular variation has been sufficiently sampled, and the paleomagnetic pole 84.1°N, 171.2°E (A_{95} = 8.6°, K = 29, N = 11) represents a time-averaged field. Nevertheless, because of the limited number of sites, the paleosecular variation estimate and the paleomagnetic pole are somewhat imprecise.

Comparing this new pole with the six previously published Miocene paleomagnetic poles for Massif Central [*Roche*, 1951, 1960; *Prévot*, 1963; *Wensink* 1970; *Biquand and Prévot*, 1972; *Prévot*, 1975], some differences are found. Two of the studies [*Prévot*, 1963, 1975] deal with Velay Oriental and partly cover the sites sampled in this study. Direct comparison of the mean directions of the five lava flows studied by both us and *Prévot* [1963, 1975] reveals a difference of 2° to 10° except for a viscous and reversely magnetized lava flow (site 3, Table 1) where it reaches 36°. We suspect that these differences are due to the inadequacy of AF demagnetization, which was the only technique used in the earlier studies, for completely erasing viscous remanent magnetization (VRM) in some volcanic rocks [*Prévot*, 1981]. Including the older results in the calculation of a global paleomagnetic pole could therefore result in a systematic bias, and we choose not to do it. We do, however, find the same magnetic polarity, as originally suggested in the extensive study of *Prévot* [1975]. The high reversal rate (~ 5 Myr^{-1}) makes a correlation of the Velay Oriental magnetozones to the geomagnetic polarity timescale [*Cande and Kent*, 1995] impossible.

The pole obtained in this study is shown in Figure 6 together with the synthetic apparent polar wander path (APWP) for Eurasia [*Besse et al.*, 1996]. Owing to the limited number of sites, the 95% confidence limit is large and covers the entire 0 to 80 Ma part of the APWP; the mean, however, is in agreement with the 10 Ma APWP point. The pole is therefore in support of the suggested stable tectonic setting of Velay Oriental [*Mergoil et al.*, 1993] and further indicates a dipole structure of the time-averaged field.

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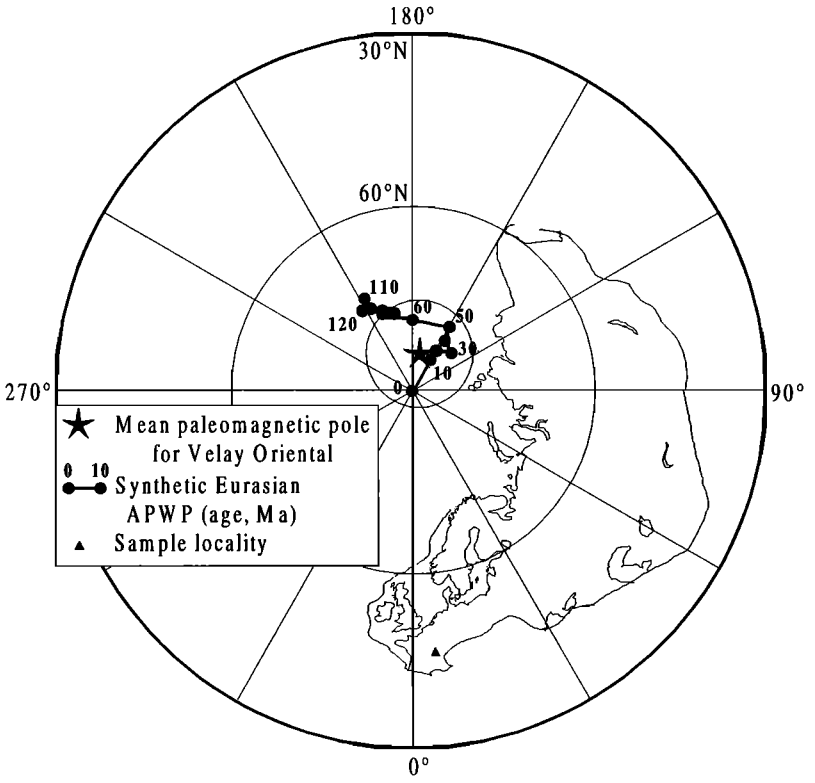


Figure 6. Paleomagnetic pole for Velay Oriental with Fisher 95% confidence limit. Also shown is the synthetic apparent polar wonder path (APWP) for Eurasia [*Besse et al.*, 1996].

6. Paleointensity Determinations

The Thellier method in its classic form [*Thellier and Thellier*, 1959] was used for the major part of the paleointensity experiments. At each temperature step the samples were heated twice with an applied field: positive for the first heating and negative for the second. In order to detect changes in the ability of the samples to acquire a TRM, sliding pTRM checks [*Prévot et al.*, 1985] were performed after every second heating step. All heatings were made in vacuum better than 10^{-4} mbar, and remanence was measured on a JR5a spinner magnetometer.

From the group of low T_c basalts, 26 samples (nine cooling units) with almost reversible κ(T) curves were selected. While only minor changes were observed during κ(T) experiments, the majority of samples underwent drastic changes during the paleointensity experiments, evidenced by nonlinear NRM-TRM plots, large changes in room temperature susceptibility and noisy "associated-NRM" demagnetization (Figure 7b). This difference is probably related to the numerous and much longer heating/cooling cycles in the paleointensity experiments ($2^1/_2$-hour heating time and ~7 hours cooling) compared to the κ(T) measurements. Only two samples (one cooling unit) yielded usable paleointensity data. One of these samples is shown in Figure 7a.

From the high T_c basalts, 10 samples (three cooling units) were chosen. None of these samples had perfectly reversible κ(T) curves, all showing some minor changes. Nine of the samples have NRM-TRM plots characterized by a loss of NRM without TRM acquisition at moderate temperatures (Figure 8b) probably due to titanomaghemite destruction. The only accepted paleointensity estimate is shown in Figure 8a.

The baked sediments proved ideal for Thellier experiments with 11 samples out of 13 yielding reliable paleointensity estimates. All accepted samples have linear NRM-TRM plots with positive pTRM checks, no changes in susceptibility, and a single-component "associated-NRM" demagnetization (Figure 9a). Two samples from St.Clement 6 were rejected due to chemical/crystalline alterations evidenced by convex-up NRM-TRM plot and larger changes in the susceptibility values (Figure 9b).

In order to further characterize the baked sediments, four samples were analyzed by introducing an additional pTRM-tail check into the *Coe* [1967] version of the Thellier method. We use the term "pTRM-tail check" to distinguish it from the standard pTRM checks [*Thellier and Thellier*, 1959; *Prévot et al.*, 1985]. In the Coe version, at each temperature step the samples are heated twice, in zero field for the first heating and under an applied field for the second heating. In order to perform the pTRM tail check, samples are heated a third time to the same temperature in zero field. Ideally, the first and the third zero field heatings to the same temperature should result in the same remaining NRM. A difference between first and third heating could correspond to an unremoved "pTRM tail" stemming from the second in-field heating, and residing in grains with unblocking temperature higher than blocking temperature (i.e., MD grains [*Bol'shakov and Shcherbakova*, 1979]). Alternatively, a difference between first and third heating could be caused by the production of a chemical remanent magnetization residing in grains with T_{ub} higher than the temperature of the heating step. In the later case we should also observe negative standard pTRM checks, changes in the direction of the NRM remaining, and changes in room temperature susceptibility. Standard pTRM check is easily obtaind after the pTRM tail check by heating to the desired temperature and cooling in a field. Examples of Thellier experiments with pTRM tail checks are shown in Figure 10. Both samples have positive standard pTRM checks

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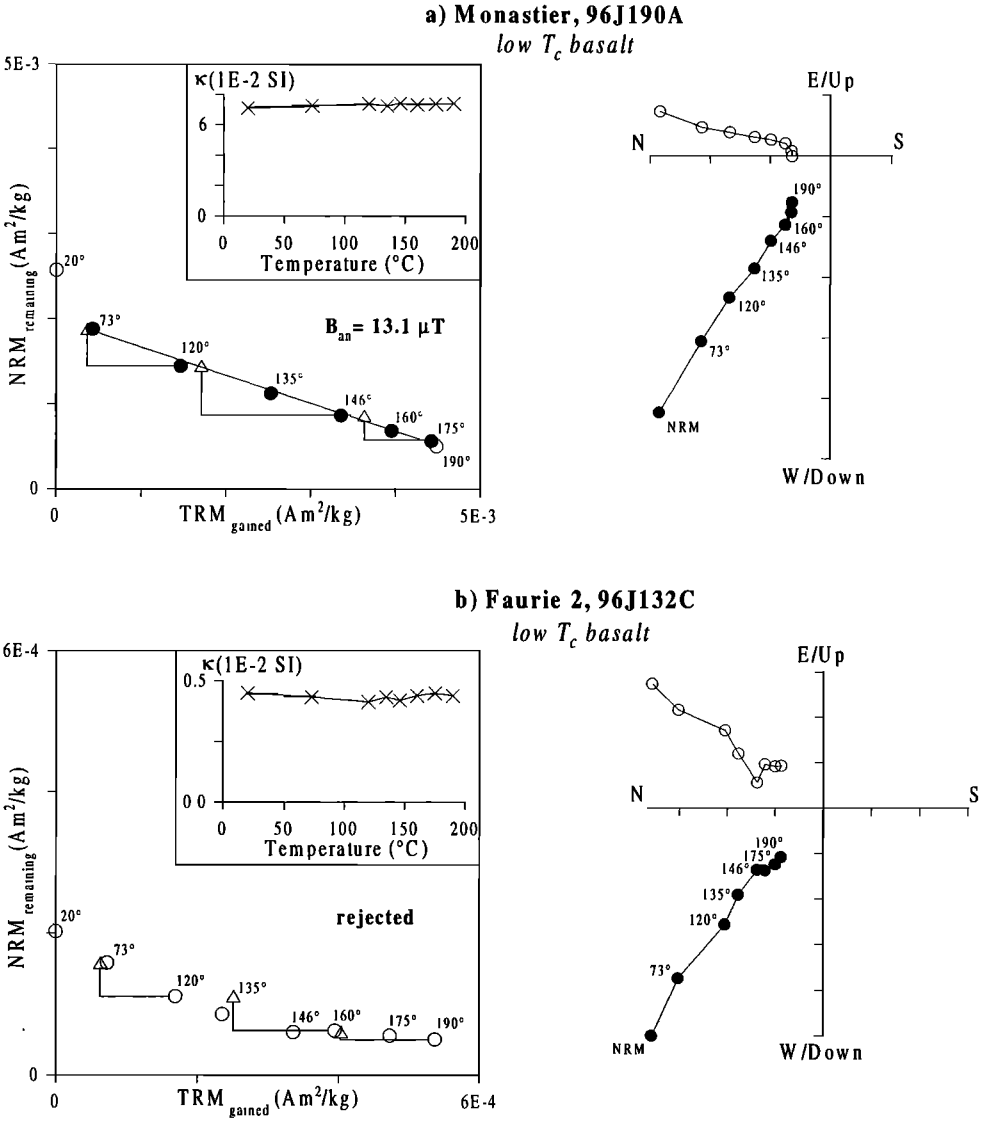


Figure 7. Examples of NRM-TRM plots and orthogonal projections of the "associated-NRM" demagnetization for the low T_c basalts. (a) Accepted sample; (b) rejected sample. Solid (open) circles on the NRM-TRM plots are accepted (rejected) points, triangles are pTRM checks. Solid (open) symbols on orthogonal projections are vertical (horizontal) planes. The susceptibility κ after each heating step is also shown.

documenting the thermal stability of baked sediments while positive pTRM tail checks ensure the SD character of the remanence carriers.

The results of all accepted paleointensity estimates and corresponding virtual dipole moments (VDMs) are summarized in Table 3.

7. Discussion

The 90% failure rate that we find in our study of both high and low T_c basalts is unfortunately not exceptional in Thellier paleointensity studies on lava flows and dikes. The dominant reason for these discouraging high failure rates is alteration of the magnetic properties during laboratory heatings. Out of 36 preselected basaltic samples, we rejected 33 due to obvious alterations (e.g., Figures 7b and 8b). The baked sediments proved much more stable and only two out of 13 samples were rejected due to alteration (Figure 9b).

Even for the three accepted basaltic samples, minor alterations seem to be taking place, and some questions can be raised regarding the reliability of these paleointensity estimates. For example, for the low T_c basalt sample shown in Figure 7a, a minor, but systematic deviation of the pTRM checks, could indicate a gradual change in pTRM capacity, although NRM-TRM points fall on a straight line. The methods of *Valet et al.* [1996] and *McClelland and Briden* [1996], designed to correct the effect of alteration during Thellier paleointensity experiments, assume that chemical changes only involve grains with blocking temperatures lower than the last heating step. These methods are therefore not applicable to the low T_c basalts (Figure 7) for which the alteration probably is related to oxyexsolution producing more Ti-poor titanomagnetites with higher Curie and blocking temperatures. The alterations observed in the high T_c basalts (Figure 8b) probably relate to destruction of maghemite, which is a secondary mineral, and correction for the alteration would therefore be meaningless.

Alteration of the magnetic remanence carriers is in general easily detected by nonlinear NRM-TRM plots, failure of pTRM checks, changes in susceptibility and noisy "associated-NRM" demagnetization (Figures 7b, 8b, and 9b). Another less easily

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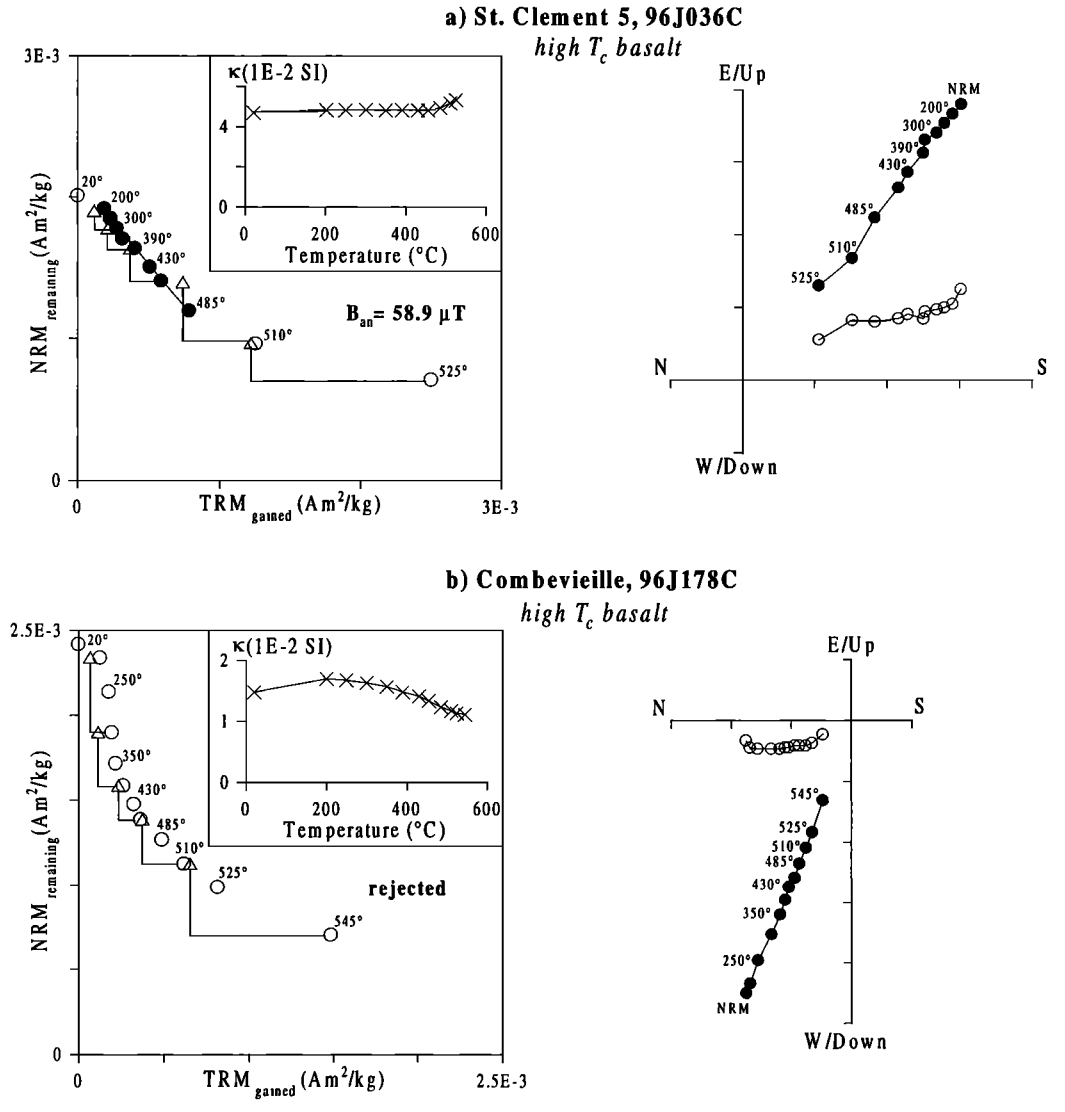


Figure 8. Examples of NRM-TRM plots together with the orthogonal projections of the "associated-NRM" demagnetization for the high T_c basalts. (a) Accepted sample; (b) rejected sample. Notation as in Figure 7.

recognized problem relates to the grain size of the magnetic remanence carriers. Only SD grains fulfill the Thellier laws of independence and additivity of pTRMs, i.e., the foundation of the Thellier paleointensity method. For MD grains the Thellier laws are not obeyed [e.g., *Bol'shakov and Shcherbakova*, 1979; *Dunlop and Özdemir*, 1997], and therefore MD material cannot be expected to give correct estimates of the paleofield strength. The concave down shape of MD material in NRM-TRM plots [*Levi*, 1977; *Xu and Dunlop*, 1995] is rarely recognized in Thellier experiments on natural rocks as alteration often leaves only the low-temperature points untouched, and a failure to discern MD material would therefore lead to an overestimate of the paleofield intensity. The indirect use of hysteresis measurements to deduce grain sizes is not always straightforward, as illustrated by the wasp-waisted hysteresis curves of the baked sediments (Figure 3c). Furthermore, it is difficult to discern from hysteresis parameters indicating a mixture of SD and MD grains to what degree the MD grains will affect the paleointensity estimate (rarely does a natural sample consist entirely of SD grains).

Introducing a pTRM tail check, we present a simple method to test the domain state of the remanence carriers during the Thellier experiment, in a way that relates directly to the behaviour of pTRMs (Figure 10). MD material can be recognized by having a "pTRM tail" with unblocking temperature higher than the blocking temperature [*Bol'shakov and Shcherbakova*, 1979; *Dunlop and Özdemir*, 1997] and therefore resulting in a negative pTRM tail check. In order to also monitor alterations, the pTRM tail check must be used in addition to the standard sliding pTRM checks [*Prévot et al.*, 1985]. For the baked sediments both checks are positive over a large fraction of the NRM-TRM spectra, indicating that the remanence resides in thermally stable SD grains. We note that this information could not be discerned from the rock magnetic experiments, as for example, hysteresis measurements hold no information about grain sizes due to the mixing of two to three separate coercivity populations. Likewise, the thermal stability of the remanence carrying grains could not be inferred from the perfectly reversible κ(T) curves (Figure 3c) as susceptibility is controlled by the SP grains.

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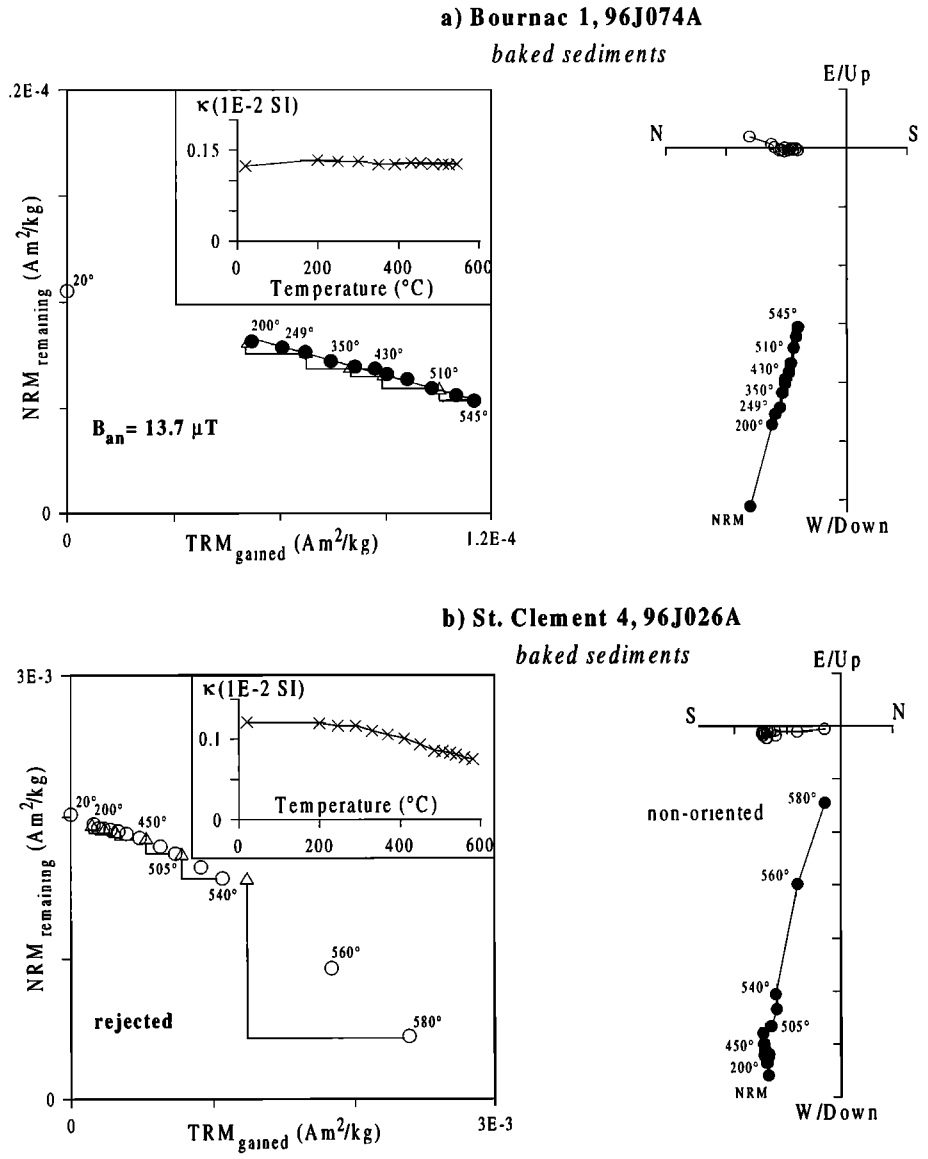


Figure 9. Examples of NRM-TRM plots together with the orthogonal projections of the "associated-NRM" demagnetization for baked sediments. (a) Accepted sample; (b) rejected sample. Notation as in Figure 7.

However, even in the case of a remanence residing in thermally stable SD grains, a still fundamental question regarding the nature of the NRM must be answered, as obviously, the Thellier method is only valid when the NRM is a TRM. The major question regarding the suitability of baked sediments for Thellier experiments should therefore be whether or not the NRM is a TRM? Unfortunately, there is no method to distinguish between TRM and chemical remanent magnetization (CRM), and hence no forthright answer can be given. The directional results from baked sediments and overlying lava flows agree well, indicating that the remanence was acquired at the time of the baking. Unfortunately, none of the samples from overlying flows yielded acceptable paleointensity estimates. The thermal stability of the baked sediments in the laboratory heatings indicates that the chemical reactions producing the magnetic minerals must have gone to completion during the original cooling of the overlying lava. Moreover, no typical products of weathering in baked sediments (i.e., goetite or lepidocrocite), which can produce large deviations from linearity during Thellier experiments [*Barbetti and Flude*, 1979], were observed in the samples. Then the question would be whether or not the reaction was ended above the Curie temperature. On theoretical ground, *McClelland* [1996] suggested that grain-growth CRM carried by SD magnetite or hematite grains can be distinguished from a TRM by nonlinear NRM-TRM plots above 400°C. We observe a linear relationship between NRM and TRM well above 400°C, which according to the theory of McClelland suggests a TRM. We therefore assume that the paleointensity estimates that we derive from the baked sediments provide reliable information on past geomagnetic field.

8. Strength of the Miocene Dipole Moment

A multitude of factors can lead to erroneous estimates of the paleofield intensity. Owing to the poor documentation of many previously published paleointensity determinations it is often hard to assess the reliability of the results. In our analysis of Miocene paleointensity data we chose to follow the reliability

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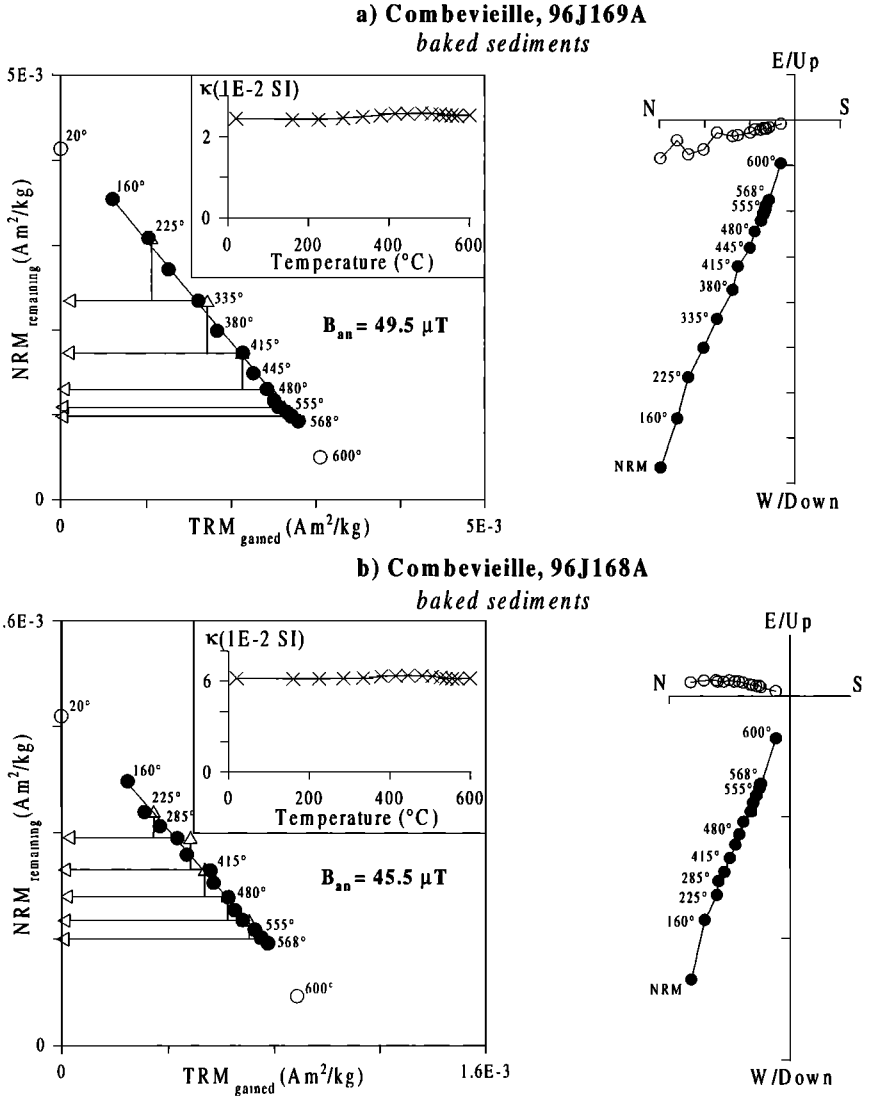


Figure 10. Examples of NRM-TRM plots with pTRM tail checks. The pTRM tail checks represent the difference between the remanence remaing after first and third heating to the same temperature in zero field (see text for further details). The difference is plotted on the TRM axis with a triangle. Sliding pTRM checks are performed after pTRM tail check. Notation as in the Figure 7.

criteria of *Perrin and Shcherbakov* [1997], which simply demand internal consistency of the paleofield provided by different specimens from the same cooling unit, i.e., at least three specimens per cooling unit, a maximum standard deviation on the paleointensity estimate <10%, and no transitional data. Only two of our estimates, from baked sediments from Bournac 1 and Combevieille, fulfill this criteria (Figure 11 and Table 3). From the 200 Miocene determinations available in the paleointensity database [*Perrin et al.*, 1998], which records all published paleointensity results, only 10 determinations sampled at Steens Mountain [*Prévot et al.*, 1985] fulfill the reliability criteria (Figure 11). All results in the database also correspond to single cooling units and therefore represent instant readings of the total Earth magnetic field. In addition to these data, 10 Miocene paleointensity estimates have recently been obtained from studies on submarine basaltic glasses (SBG) [*Juárez et al.*, 1998]. As the SBG are unoriented, virtual axial dipole moments are shown in Figure 11. However, the SBG results correspond to mean values obtained from a given drill hole, thus partly averaging the nondipole terms of the magnetic field, and cannot be compared directly to the other results.

Because of the extremely limited number of data and the different influence of secular variation between both groups of estimates, the calculation of a mean Miocene virtual dipole moment could be very strongly biased. Even though presently available results point toward a field much lower for the Miocene than present-day VDM (8.0×10^{22} A m^2), many more results are needed to define the dipole intensity of the geomagnetic field during Miocene.

9. Conclusions

From a paleomagnetic study of Miocene volcanic rocks and baked sediments sampled at Velay Oriental, we obtain a paleomagnetic pole with coordinates 84.1°N, 171.2°E (A_{95} = 8.6°, K = 29, N = 11). The pole is supported by a positive reversal test

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Table 3. Velay Oriental Paleointensity Determinations

Site	Specimen	Rock Type	ΔT,°C	n	f	g	q	$B_a \pm \sigma$, μT	VDM, 10^{22} A m^2
St. Clement 6	96J029B	baked sediments	200 - 560	12	0.55	0.89	25.5	17.5 ± 0.3	2.9
St. Clement 7	96J036C	high T_c basalt	200 - 485	8	0.34	0.82	9.6	58.9 ± 1.7	11.2
Bournac 1	96J074A	baked sediments	200 - 545	11	0.27	0.89	7.9	13.7 ± 0.4	2.2
	96J075A	baked sediments	250 - 525	9	0.33	0.87	12.4	13.5 ± 0.3	2.2
	96J077A	baked sediments	200 - 545	11	0.31	0.89	7.5	12.1 ± 0.4	1 9
	96J078A	baked sediments	200 - 580	13	0.50	0.89	10.1	12.1 ± 0.5	1.9
Mean Bournac 1								12.9 ± 0.9	2.0 ± 0.1
Combevieille	96J166A	baked sediments	200 - 545	11	0.51	0.88	31.8	56.3 ± 0.8	8.4
	96J168A	baked sediments	160 - 568	13	0.48	0.90	17.5	45.5 ± 1.1	6.8
	96J169A	baked sediments	160 - 568	13	0.62	0.88	36.3	49.5 ± 0.7	7.4
	96J170A	baked sediments	200 - 545	11	0.41	0.89	30.9	48.3 ± 0.6	7.2
	96J172A	baked sediments	200 - 545	11	0.31	0.90	17.7	45.6 ± 0.7	6.8
	96J173A	baked sediments	225 - 555	11	0.30	0.89	8.7	41.0 ± 1.3	6.1
Mean Combevieille								47.7 ± 4.7	7.3 ± 0.7
Monastier	96J190A	low T_c basalt	73 - 175	6	0.66	0.78	14.5	13.1 ± 0.5	2.4
	96J195C	low T_c basalt	73 - 175	6	0.67	0.70	12.6	16.4 ± 0.6	2.9
Mean Monastier								14.7 ± 1.7	2.7 ± 0.4

ΔT is the temperature interval used for paleointensity determination; n is the number of accepted points; f, g, and q are the NRM fraction, the gap factor, and the quality factor, respectively [*Coe et al.*, 1978]; $B_a \pm \sigma$ is the sample paleointensity with the associated error, and VDM is virtual dipole moment.

and corresponds well with the apparent polar wander path for Eurasia. Moreover, five new $^{40}Ar/^{39}Ar$ plateau ages falling between 9.20 and 13.55 Ma were obtained.

High-quality paleointensity estimates were obtained from baked sediments coming from two seperate cooling units with VDMs 2.0 ± 0.1 and 7.3 ± 0.7 × 10^{22} A m^2. A selection of reliable Miocene results points toward Miocene VDM significantly lower than the present-day dipole moment; however, the data set is not large enogh to average secular variation and calculate dipole field intensity in Miocene. Perhaps the most striking feature in the Miocene paleointensity database (Figure 11) is that only 5% of the data fulfill the proposed reliability criteria.

In addition to the pTRM check originally suggested by *Thellier and Thellier* [1959] and modified by *Prévot et al.* [1985], we suggest introducing a second pTRM tail check, designed to test the domain state of the remanence carriers. Combining the two pTRM checks it was shown that in the case of baked sediments the remanence is carried by thermally stable SD grains.

Baked sediments have long been known as good recorders of the paleofield [*Bruhnes*, 1906], and seemingly reliable paleointensity estimates have previously been obtained [e.g., *Barbetti and Flude*, 1979]. The availability of baked sediments holds great potential for a better distribution of high-quality paleointensity data in time.

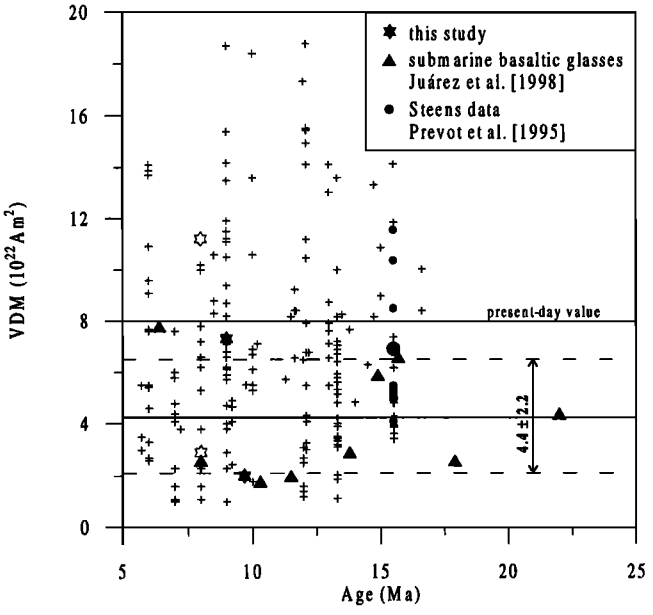


Figure 11. Evolution of virtual dipole moments for Miocene. Determinations fulfilling selection criteria are shown by solid symbols (stars if obtained in this study). Circles are VDMs of 10 selected Steens estimates [*Prévot et al.*, 1985]. Triangles are virtual axial dipole moments obtained from submarine basaltic glasses [*Juárez et al.*, 1998]. Determinations failing to fulfill selection criteria are shown by crosses (stars if obtained in this study).

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